

## Lecture 11: Priors as Fake Data

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## 11.1 How Much Should We Believe a 3–0 Start?

Suppose an MLB team starts the season with three wins and no losses. What should we use as its underlying win probability  $p$ , the probability the team would win a randomly selected game under the simplified model?

- 1.000: use the observed record.
- 0.500: ignore three noisy games.
- Something between: combine the record with outside information.
- Something team-specific: a preseason favorite should start above 0.500.

An estimate based on little data can move dramatically because of chance. When useful information exists before these three games, estimation can combine that information with the observed data. How should we do this?

First, let  $W$  and  $L$  be a team's observed wins and losses, with  $m = W + L$ . Under a simplified model,

$$W \mid p \sim \text{Binomial}(m, p).$$

The maximum likelihood estimate of  $p$  is

$$\hat{p}_{\text{MLE}} = \frac{W}{W + L}.$$

This estimate is very sensible with enough games, but clearly extreme and unstable early in the season.

**Adding information to the MLE.** One way to stabilize a noisy binomial proportion is to add a small amount of information before taking the ratio. The Agresti–Coull interval uses  $\tilde{p} = (W + 2)/(W + L + 4)$ , which acts like adding two wins and two losses. More generally, adding  $a$  fake wins and  $b$  fake losses gives

$$\tilde{p}_{a,b} = \frac{W + a}{W + L + a + b}.$$

This forces two questions into the open: where is the added information centered,  $a/(a + b)$ , and how strong is it,  $a + b$ ? For a 3–0 start, adding 15 fake wins and 15 fake losses gives  $18/33 \approx 0.545$  instead of 1.000. Bayesian updating turns this fake-data intuition into a probability model.

## 11.2 Bayes' Rule

Bayesian updating has three pieces: the **prior**  $\mathbb{P}(p)$  describes belief before the games, the **likelihood**  $\mathbb{P}(W, L \mid p)$  describes the observed record as a function of  $p$ , and the **posterior**  $\mathbb{P}(p \mid W, L)$  is the updated distribution after seeing the games.

Bayes' rule combines them:

$$\begin{aligned}\mathbb{P}(p \mid W, L) &= \frac{\mathbb{P}(W, L \mid p)\mathbb{P}(p)}{\int_0^1 \mathbb{P}(W, L \mid u)\mathbb{P}(u) du} \\ &\propto \mathbb{P}(W, L \mid p)\mathbb{P}(p).\end{aligned}$$

The denominator does not depend on  $p$ ; it only makes the posterior integrate to one.

For binomial data, a convenient prior is

$$p \sim \text{Beta}(\alpha, \beta).$$

The two parameters determine where the prior distribution is centered and how concentrated it is. Define  $m = \alpha/(\alpha + \beta)$  and  $s = \alpha + \beta$ . For a Beta prior,

$$\mathbb{E}[p] = m, \quad \text{Var}(p) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{m(1 - m)}{s + 1}.$$

The mean  $m$  locates the prior belief: the value of  $p$  that seems typical before seeing the team's new games. The variance describes prior uncertainty around that center. Holding  $m$  fixed, larger  $s$  puts more prior mass near  $m$  and leaves less room for extreme values.

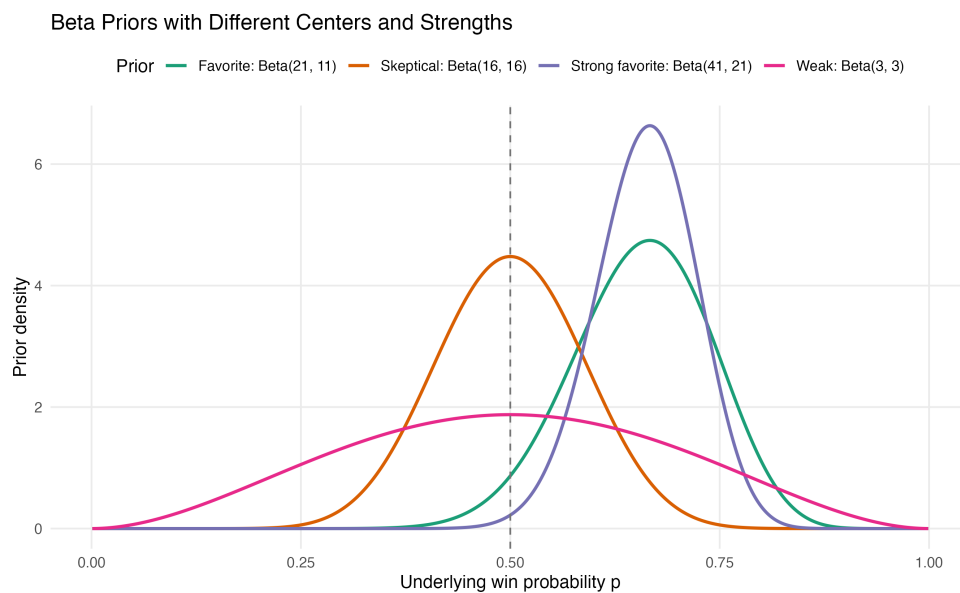


Figure 11.1: Beta prior densities with different centers and strengths.

A prior is a quantitative modeling assumption. In sports, its center might come from league history, roster information, or a market rating. Its strength should reflect how much confidence we place in that information.

**Example 11.1** (Turning a Market Projection Into a Prior). Suppose a betting market projects a team for 88 wins in 162 games. The implied prior center is

$$m = \frac{88}{162} = 0.543.$$

If we judge that projection to be worth  $s = 40$  games of evidence, set

$$\alpha = sm, \quad \beta = s(1 - m).$$

Changing  $s$  from 10 to 40 to 100 changes the prior's influence without changing its center. Prior elicitation translates substantive information into a center and strength. Prior choice depends on the analyst's pre-data information and judgment, and it can materially change posterior conclusions.

### 11.3 Beta-Binomial Updating

For the Beta-Binomial model,

$$\begin{aligned}\mathbb{P}(W, L | p) &\propto p^W (1 - p)^L, \\ \mathbb{P}(p) &\propto p^{\alpha-1} (1 - p)^{\beta-1}.\end{aligned}$$

Multiplying the two kernels gives

$$p | W, L \sim \text{Beta}(\alpha + W, \beta + L).$$

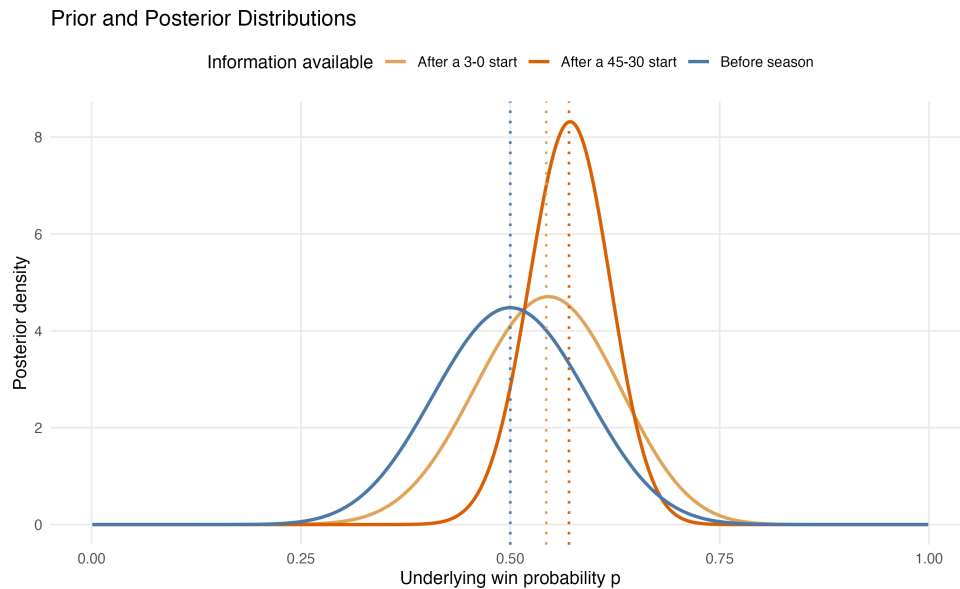


Figure 11.2: A Beta(16,16) prior and posterior distributions after 3 and 75 games.

This update is a general Bayesian pattern:

$$\text{posterior information} = \text{prior information} + \text{information from the data}.$$

Updating can also happen sequentially. After one batch of games, the posterior

$$\text{Beta}(\alpha + W_1, \beta + L_1)$$

becomes the prior for the next batch. Combining it with  $(W_2, L_2)$  gives

$$p | W_1, L_1, W_2, L_2 \sim \text{Beta}(\alpha + W_1 + W_2, \beta + L_1 + L_2).$$

The order of the batches does not matter under this model. The posterior is the accumulated state of information.

## 11.4 Posterior Mean, MAP, and Fake Data

Once we have a posterior distribution, we still need to decide what summary answers the question.

### 11.4.1 Posterior Mean

Under squared-error loss, the optimal point estimate is the posterior mean:

$$\hat{p}_{\text{mean}} = \mathbb{E}[p \mid W, L] = \frac{W + \alpha}{W + L + \alpha + \beta}.$$

This looks like adding  $\alpha$  pseudo-wins and  $\beta$  pseudo-losses.

### 11.4.2 Posterior Mode or MAP

The maximum a-posteriori estimate is the mode of the posterior:

$$\hat{p}_{\text{MAP}} = \frac{W + \alpha - 1}{W + L + \alpha + \beta - 2},$$

when  $\alpha + W > 1$  and  $\beta + L > 1$ . The MAP therefore looks like adding  $\alpha - 1$  fake wins and  $\beta - 1$  fake losses.

**Example 11.2** (A 3–0 Start). With  $p \sim \text{Beta}(16, 16)$  and a 3–0 start,

$$\begin{aligned}\hat{p}_{\text{MLE}} &= 1.000, \\ \hat{p}_{\text{mean}} &= \frac{19}{35} = 0.543, \\ \hat{p}_{\text{MAP}} &= \frac{18}{33} = 0.545.\end{aligned}$$

The posterior mean and MAP are similar here, but they answer different optimization questions.

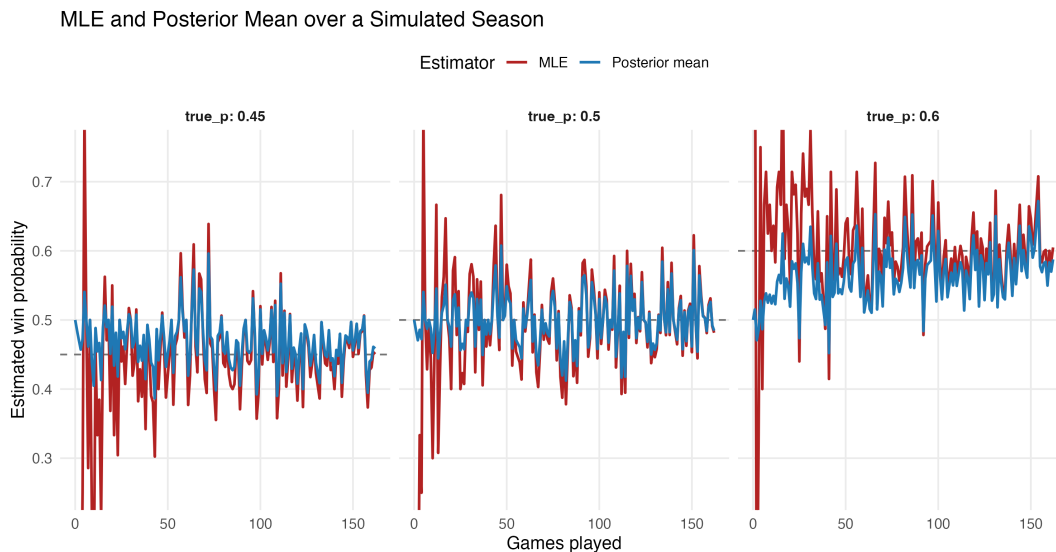


Figure 11.3: MLE and posterior mean over one simulated season.

## 11.5 Posterior Inference and Prediction

The posterior directly supports probability statements. A 95% credible interval is

$$[q_{0.025}\{\text{Beta}(\alpha + W, \beta + L)\}, q_{0.975}\{\text{Beta}(\alpha + W, \beta + L)\}].$$

Conditional on the model and prior, there is 95% posterior probability that  $p$  lies inside this interval.

If  $W_{\text{future}}$  is the number of wins in the next  $r$  games, then

$$W_{\text{future}} \mid p \sim \text{Binomial}(r, p).$$

Integrating over the posterior for  $p$  gives

$$W_{\text{future}} \mid W, L \sim \text{Beta-Binomial}(r, \alpha + W, \beta + L),$$

$$\mathbb{E}[W_{\text{future}} \mid W, L] = r \frac{\alpha + W}{\alpha + \beta + W + L}.$$

Posterior prediction averages future-game randomness over uncertainty in  $p$ . It is wider than uncertainty about  $p$  alone because it includes both parameter uncertainty and game-to-game randomness. A forecast of future wins is not the same as an estimate of  $p$ . Even if  $p$  were known, baseball games would still be random.

## 11.6 Prior Sensitivity

A prior can improve estimation, but it can also mislead. Prior sensitivity examines:

1. What information determined the prior center?
2. What does the prior strength mean in the units of this problem?
3. Would conclusions change under other reasonable priors?
4. Does the prior generate plausible data before the real data are used?

Priors are one form of **regularization**: they deliberately pull unstable estimates toward plausible values. The benefit is lower variance. The cost is bias if the target or strength is poorly chosen.

## 11.7 Takeaways

- Small samples produce extreme estimates; a prior controls how strongly to pull them back.
- A prior has a center and a strength, both of which require substantive justification.
- In the Beta-Binomial model, the posterior is  $\text{Beta}(\alpha + W, \beta + L)$ .
- The posterior mean and MAP are distinct summaries with distinct fake-data interpretations.
- Posterior distributions support intervals, predictions, and decisions, not only point estimates.

## Appendix

### A Binomial Maximum Likelihood

The Binomial likelihood kernel is

$$\mathcal{L}(p) \propto p^W (1-p)^L.$$

When  $W > 0$  and  $L > 0$ , its log is  $\ell(p) = W \log p + L \log(1-p)$ . Setting

$$\ell'(p) = \frac{W}{p} - \frac{L}{1-p} = 0$$

gives  $\hat{p}_{\text{MLE}} = W/(W+L)$ . If all games are wins or all games are losses, the maximum occurs at the boundary  $p = 1$  or  $p = 0$ , so the same formula still gives the MLE. Bayes does not discard this likelihood; it combines the likelihood with additional information encoded by the prior.

### B Suggested Reading

- Andrew Gelman et al., *Bayesian Data Analysis*, on Bayesian updating and posterior prediction.
- Alan Agresti and Brent Coull, “Approximate Is Better than Exact for Interval Estimation of Binomial Proportions,” on a frequentist fake-data connection.